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A multi-level visualization method for IT system structure

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Abstract

This article is to solve the problems of disordered and complex architectural relationships in IT in the multi-level visualization of the IT system structure, and the relationship is not transparent and intuitive. 3D visualization has its unique display advantages. By integrating the visualization concept into the IT architecture management process, Establish a multi-level visualization model of IT system structure, covering from the IT infrastructure level to the system application and business logic level, analyze location attributes, node types, rule attributes and relationship attributes, and present them in a three-dimensional visualization form to make IT internal structural relationships And the relationship between nodes is clear and intuitive. This article uses different layout algorithms such as hierarchical layout algorithm, circular distribution algorithm, star layout algorithm, etc. According to the different characteristics of the number of IT system nodes and node relationships, matching the corresponding layout algorithm makes the multi-level IT system visualization architecture more flexible. The experimental results show that the method proposed in this paper can show the IT system structure simply and clearly, and match various complex business relationships more flexibly.

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1. Introduction

In recent years, with the rapid growth of application systems and the continuous upgrading of system architecture, IT systems have become increasingly complex. In the actual IT system management work, there is a practical problem of opaque IT architecture information. It is often faced with such problems as the system architecture is changeable, the access relationship between systems is not clear, and the troubleshooting is not timely. Therefore, the research on end-to-end multi-level visualization of IT system structure is getting more and more attention, and it is regarded as one of the important problems facing modern IT system operation and maintenance management, and it is an important means to solve IT structure information and opacity.

At present, there are complex relationships between nodes in the IT system structure, such as including, initiating, relying on, belonging to, independent relationships, etc. the effective layout algorithm can help users clearly and intuitively understand the node layout in the scene, and directly control the local and global state of the scene, which is convenient for managers to control the overall system. The existing layout algorithms mainly include hierarchical layout algorithm, ring layout algorithm and star layout algorithm. Among them, the hierarchical layout algorithm can intuitively show the level of IT architecture, which is convenient for users to visually view the relationship between levels. However, when the number of nodes is large, the number of nodes in different levels is not evenly distributed. Cao Ruibin¹ used the spiral transformation method to realize the hierarchical layout algorithm of IT system structure, avoiding the top heavy problem caused by too many nodes in the hierarchical layout. Traditional 3D visual layout algorithm has the problems of unclear representation of complex concept lattice structure and unsightly graphics. Shen Xiaojiong¹¹ improved the layout algorithm by adding virtual nodes, but the method based on probabilistic graph has the problems of long running time and unstable map layout.

At present, the methods proposed above all solve the targeted layout, but when faced with complex and changeable models, they cannot visually display the business relationship layout more clearly and effectively, and there are limitations. Therefore, this article uses a variety of layout algorithms—star layout algorithm, hierarchical layout algorithm, and circular layout algorithm to effectively solve the clear and intuitive layout of the IT architecture visualization topology and improve user interaction experience.

2. Relevant Knowledge

This article proposes the overall process of a multi-level visualization method for IT system structure, which mainly includes multi-level data definition for IT system structure, multi-level visualization model of IT system structure, three-dimensional multi-level visualization and user interaction response. The process is as follows: first obtain the original data of the IT system structure, preprocess the original data of the IT system structure, and define the data level and data node according to the attributes of the source node and the level to which it belongs, and obtain the level used to map the source node and contain the corresponding The level of and the nodes containing node attributes; Match the corresponding layout algorithm according to the node attributes, obtain the association relationship between the node data according to the matched layout algorithm, and complete the data layout for each node in the display hierarchy; after the collision detection and adjustment of the completed layout node data, the adjusted The node data is visualized at multiple levels through a 3D visualization engine. This article uses a mixed layout algorithm. Effectively solve the problem of clear layout in the visual topology, and users can have an intuitive visual experience. As shown in Figure 1:

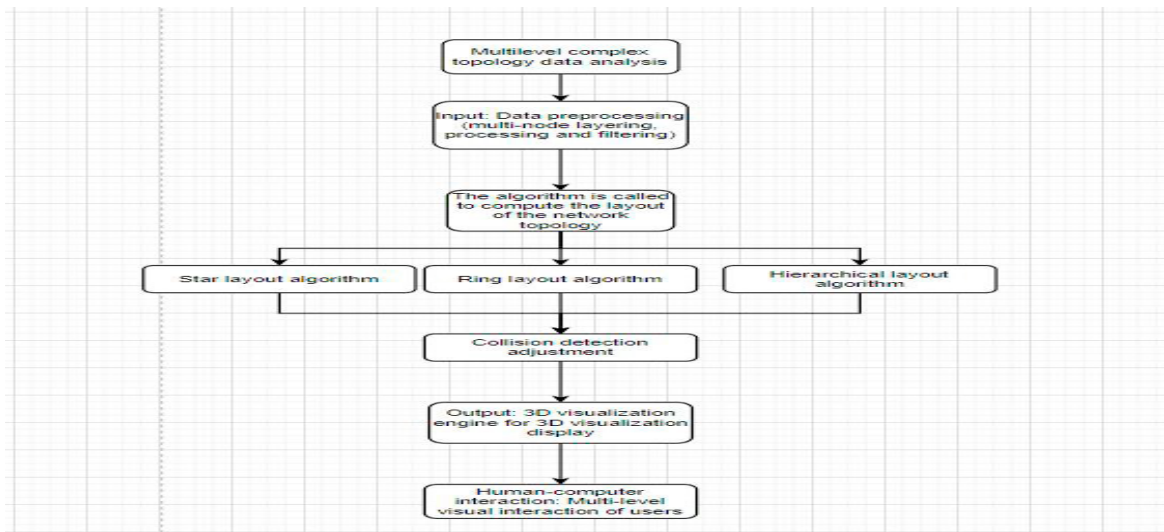


Fig. 1. Overall flow chart of this method

2.1. Multi-level visualization model of IT system structure

The multi-level visualization model of IT system structure consists of three parts: IT system structure visualization relation model, IT system structure visualization data processing model and IT system structure visualization engine rendering. Among them, the IT system structure visualization relationship model sorts out and analyzes complex business data, and establishes the relationship between each level and its node collection; the IT system structure visualization data processing model is responsible for processing and transforming the data collection with the structure of the association relationship and can be used in IT. The system structure visualization engine renders the analyzed and rendered data collection; the IT system structure visualization engine is responsible for the visualization rendering.

1) Visual relation model of IT system structure: The visual relationship model of the IT system structure is developed around specific business entities, analyzes the characteristics and attributes of each business and establishes the relationship between different business entities, so as to be managed at different levels. The set of business entities can be divided into business application layer entities (A), system platform layer entities (B), IT infrastructure layer entities (C), and computer room dynamic environment layer entities (C) for definition description. Among them, business application layer entities include secondary entities, such as access channels, firewalls, application systems, etc.; system platform layer entities, including secondary entities including web servers, databases, etc.; IT infrastructure layer entities include secondary entities and servers, Network, etc; the physical layer of the engine room dynamic ring includes the secondary entities with data centers and their locations. Define L to represent level, then $L = \{A, B, C, D, E, \dots\}$;

$A \in (\text{Access, firewall, load, encryption machine, application system} \dots)$;

$B \in (\text{web server, application server, interface platform, external interface, front-end processor, gateway server, database} \dots)$;

$C \in (\text{server, storage, network} \dots)$;

$D \in (\text{data center})$;

Based on the hierarchical relationship, this paper analyzes the internal nodes and the relationship between nodes according to the business relevance, so as to establish the association relationship. For example, the database in the system entity and the storage in the infrastructure entity constitute an association relationship.

2) Visual data processing model: On the basis of the multi-level visualization model of the IT system structure, by extracting the effective data of each data item, layering, processing, filtering, and establishing the rules, location, layout, quantity, visual rendering, related business items, constraint conditions, etc. of the node attributes Structured

processing provides support for subsequent visual rendering. This paper uses JSON string to define and transfer data structure: [hierarchy, node, relation, constraints, layout strategies, layout algorithms, layout rules, wiring rules, rendmethod, location, level, angle...]. By encapsulating the JSON format string data, filtering the node attributes in the JSON string, performing data analysis and filtering operations on the complex multi-layer topology, and for the input preprocessed data (multi-node layering, processing, filtering), then the algorithm is called to calculate the layout of the network topology, and the nodes are associated and displayed according to the constraints in the node attributes, layout strategies, layout algorithms, and layout rules, so that the visualized architecture model of the IT system structure is displayed clearly and flexibly. As shown in Figure 2.

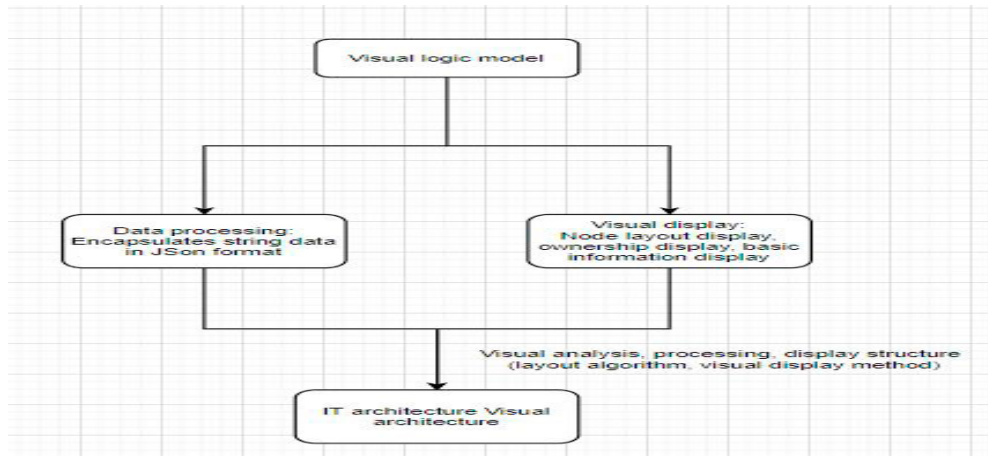


Fig. 2. Visualization transformation process of IT system structure

3. Layout algorithm flow based on visual architecture

The layout and visualization method for multi tier IT system structure is characterized by the following steps:

a) Obtain the original data of the IT system structure, the original data including the attributes of the source node of the IT system structure and the level to which it belongs;

The attributes of the source node include equipment type information, hierarchical type information, equipment relationship information, and equipment location information, and the node attributes include: the hierarchy to which they belong, and constraint conditions for matching layout algorithms,

a) Obtain the original data of the IT system structure, the original data including the attributes of the source node of the IT system structure and the level to which it belongs;

The attributes of the source node include equipment type information, hierarchical type information, equipment relationship information, and equipment location information, and the node attributes include: the hierarchy to which they belong, and constraint conditions for matching layout algorithms, Display the length of the hierarchy, user terminal icon, external system icon, access channel icon, firewall icon, web server icon, location information, node relationship data; the hierarchy to which the source node belongs includes application layer, physical layer and business layer;

b) Preprocess the original data of the IT system structure, perform data level definition and data node definition according to the attributes of the source node and the level to which it belongs, and obtain the level used to map the level of the source node and contain the corresponding application layer, physical layer, and business layer, and Nodes containing node attributes;

The data layer definition is used to set and display at least one of the application layer, the physical layer, and the business layer of the IT system structure; the data node definition is used to set the nodes in the display layer;

c) Match the corresponding layout algorithm according to the node attributes, obtain the association relationship between the node data according to the matched layout algorithm, and complete the data layout for displaying the nodes in the hierarchy;

d) After the collision detection adjustment is performed on the node data of the completed layout, the adjusted node data is visualized at multiple levels through a 3D visualization engine.

Specifically, if there is an association relationship between multiple nodes, the hierarchical layout algorithm is executed to realize the connection of multiple associated nodes; otherwise, the hybrid layout algorithm is executed.

3.1. Layout algorithm based on hierarchy

The hierarchical structure has obvious function hierarchy characteristics, and the topological structure formed has a distinct sense of hierarchy, which clearly and intuitively shows the relationship between nodes. Through the visualization model of IT system structure, the node and node relationship data of different levels are analyzed, and the layout status of all application nodes in the information system is displayed on one level. The connection mode between application nodes is displayed by connecting lines, so that managers can have an overall grasp of the overall layout and relationship of the system. The hierarchical layout algorithm is used to show the association and flow relationship between layers.

Algorithm Description:

Step 1: Obtain node data, analyze the node data to obtain the level information, location information, and node relationship data in the node attributes;

Step 2: Divide each node according to hierarchy;

Step 3: Connect the nodes according to the association relationship between the node data. The specific process is shown in Figure 3.

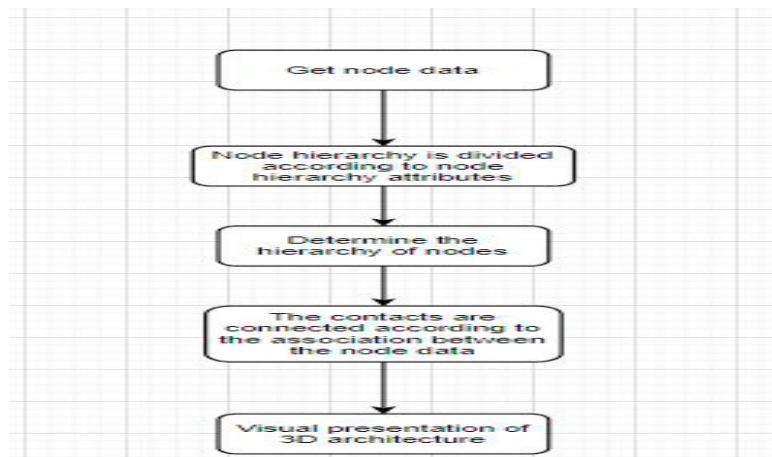


Fig. 3. Hierarchical layout algorithm flow

Obtain all the information of each node on the CPU side, and then divide it into different levels according to the data relationship (node attribute) flow direction, judge the level of the node attribute level parameters, and use spiral growth after judging the level of the node. Add nodes in the method of adding or reducing a certain distance according to the original node position, thus effectively avoiding the top-heavy situation of the hierarchical layout algorithm, and then connect the nodes and prepare three-dimensional display data for them, effectively improving The efficiency of the layout algorithm.

3.2. Layout algorithm based on star structure

Star structure is characterized by many to many topological structure relationship, which is suitable for many to many node relationship between layers. For example, a node in the application layer is regarded as the central node, and then the corresponding multiple system nodes corresponding to the node are found and regarded as the child nodes of the root node. By showing the internal structure and related relationships of application nodes, the relationship is displayed in the vertical access system structure layer in the form of connection, and the star layout algorithm is used to construct it to present the association relationship of a single application node.

Algorithm description:

Step 1: obtain the node attributes corresponding to the node data, and divide the corresponding sub levels according to the node attributes. The sub levels can dynamically increase the number of levels, and determine the central node according to the counter;

Step 2: for the divided sub-level, in a certain sub-level, the nodes are evenly distributed according to the radius distance r set by the central node, and the layout of the nodes is allocated according to the node rules; the nodes are connected with the associated nodes through the star algorithm;

Step 3: connect the nodes with association relationship to realize the star structure layout.

The determining of the central node according to the counter is specifically:

Set m nodes as $B_1 \dots B_m$, and when the number of other nodes connected to a node in m nodes exceeds the set value, set this node as the central node.

The step 2 is specifically:

Among the nodes that meet the center, they are connected according to the relationship attribute associated with the center node. The connection rule is first based on how many nodes the center node has associated with it, and then according to the number of nodes and the radius distance R , that is, each The distance from the node to the center node is assigned according to the angle; the formula is as follows:

The degree between each node: $a = 360/n$;

Among them, a is the angle formed by the arc between the node and the connected node, and n is the number of nodes.

3.3. Layout algorithm based on ring structure

The ring layout algorithm is suitable for nodes in the hierarchy with a small number of nodes. The implementation of the ring layout algorithm is as follows:

Step 1: Obtain the node attributes corresponding to the node data, and divide the corresponding sub-levels according to the node attributes. The number of sub-levels can be dynamically increased. The sub-levels are connected through the node relationship, and the next-level branch node is connected to it;

Step 2: For the divided sub-levels, in a specific sub-level, the nodes are connected to the nodes in the distributed sub-levels through a ring algorithm, so as to realize the layout of $A_1, A_2, A_3 \dots A_n$ nodes;

Step 3: Then connect a node of the first level in the sub-level to its associated branch application node of the next sub-level.

4. Experiment and analysis

The interactive response time in the process of IT system structure visualization is one of the most obvious indicators of real-time rendering efficiency of multi-level visualization scene of IT system structure. The interactive response time is mainly related to the relationship model of IT system structure visualization, the data processing model of IT system structure visualization, the rendering of IT system structure visualization engine and the algorithm of layout topology. The effect diagram of the experiment is shown in Fig 4.

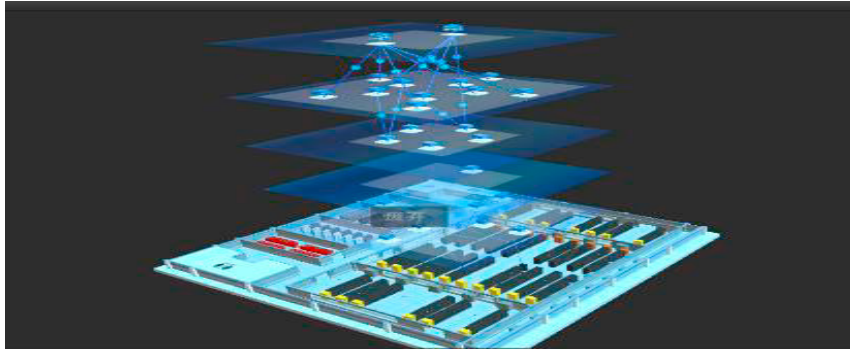


Fig. 4. Visual operation effect diagram of IT system structure.

Through the visual operation effect diagram of IT system structure, we can clearly see that the layout of nodes is more balanced, and the nodes are clearer and more beautiful by using the layout algorithm proposed in this paper. When the number of nodes is large and the number of relationships is large, the connection between nodes and between levels can still be clearly visible and beautiful.

In this paper, hierarchical layout method, star layout method and ring layout method are used to visualize the topology of IT system structure. As shown in Table 1, the performance of visualization rendering for three different topologies on the same hardware platform is shown.

Table 1. Comparison Chart of algorithm performance.

Number of nodes / frame (index)		Hierarchical layout method	Star layout method	Ring layout method
500	System response time / S	0.52	0.42	0.72
	Average frame rate / FPS	55	61	67
	Average CPU usage/	26	21	30
5000	System response time / S	1.13	0.92	1.83
	Average frame rate / FPS	43	49	53
	Average CPU utilization /%	28	23	31
20000	System response time / S	1.96	1.26	2.12
	Average frame rate / FPS	34	36	39
	Average CPU utilization /%	30	41	35

Through the analysis of Table 1, it can be seen that the algorithm proposed in this paper can maintain a high frame rate in the case of large number of nodes, and make a rapid response to the system at the same time. The experimental analysis shows that the average frame rate of the three methods can meet the actual application needs.

5. Summary and Outlook

In order to improve the multi-level visualization of IT system structure, a multi-level visualization method for IT system structure is proposed. Through the establishment of IT system structure visualization model (IT system structure visualization relation model, IT system structure visualization data processing model and IT system structure visualization engine rendering), the IT architecture internal is clearly visible. On this basis, this paper uses star layout algorithm, ring layout algorithm, hierarchical layout algorithm, according to the different characteristics of IT system structure relationship, suitable layout algorithm is applied to solve the complex and diverse difficulties of IT system structure relationship, so that IT system can be more clearly visible.

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